

NPTEL Course on
"Computer System Optimization for ML"

Lecture #53

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Agenda

- **Performance Characterization of Deep Learning Workloads (contd.)**

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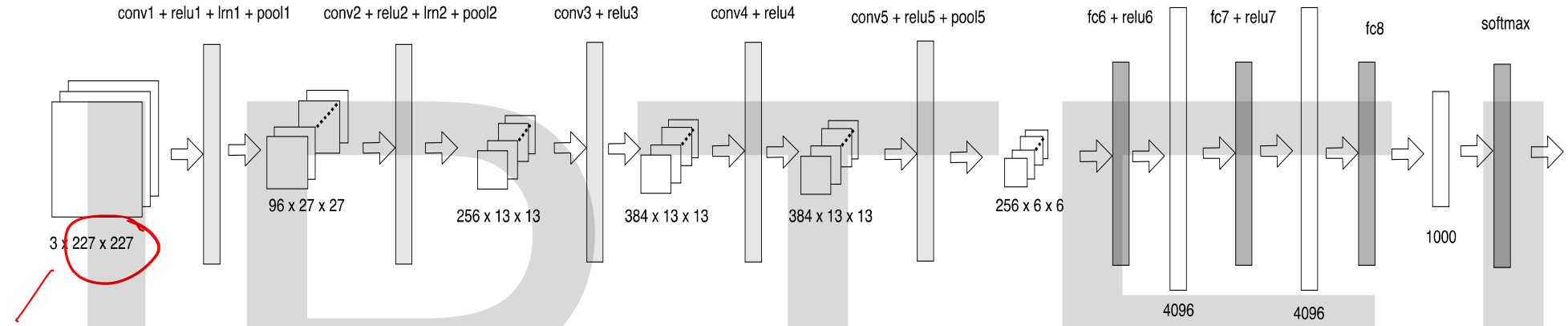
Characterization of CNN on GPUs [1]

- **How microarchitectural structures in a GPU impact the performance of CNN**
- **Uses AlexNet (only)**
- **CuDNN and CuBLAS libraries**

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[1] Dong, Shi, et al. "Characterizing the microarchitectural implications of a convolutional neural network (cnn) execution on gpus." *Proceedings of the 2018 ACM/SPEC International Conference on Performance Engineering*. 2018.

AlexNet (Reminder)



Layer	Number of Operations	Layer	Number of Operations
conv1	210M	conv4	448M
relu1	290K	relu4	65K
lrn1	4M	conv5	299M
pool1	630K	relu5	43K
conv2	896M	pool5	83K
relu2	186K	fc6	75M
lrn2	3M	relu6	4K
pool2	389K	fc7	33M
conv3	299M	relu7	4K
relu3	65K	fc8	8M
softmax	1M		

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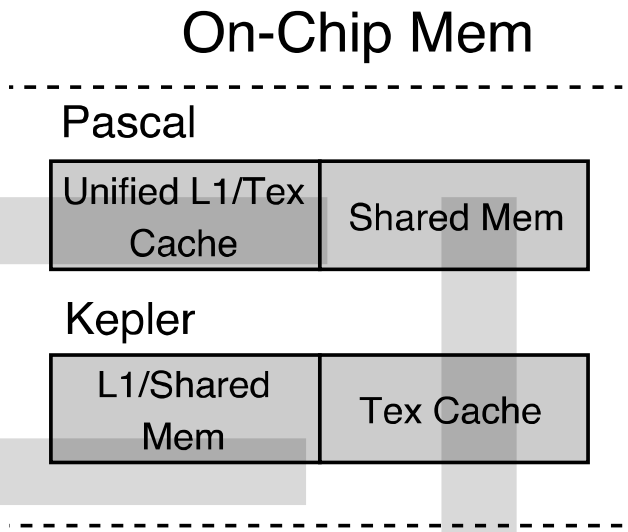
Basic Operation

- **Fully Connected**
  - Matrix multiply
- **Convolution**
  - Matrix multiply (GEMM)
  - FFT
  - Winograd (computationally efficient)
- **Non-linear operations**
  - Relu, Sigmoid, Tanh
  - Element-wise operation

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Experimental Setup

- **Two different GPU platforms**
  - GTX 1080 (Pascal family)
  - K40 (Kepler)
- **Uses nvprof (deprecated)**
  - Nsight
- **Challenge in capturing layer-specific information**
- **One iteration of AlexNet**

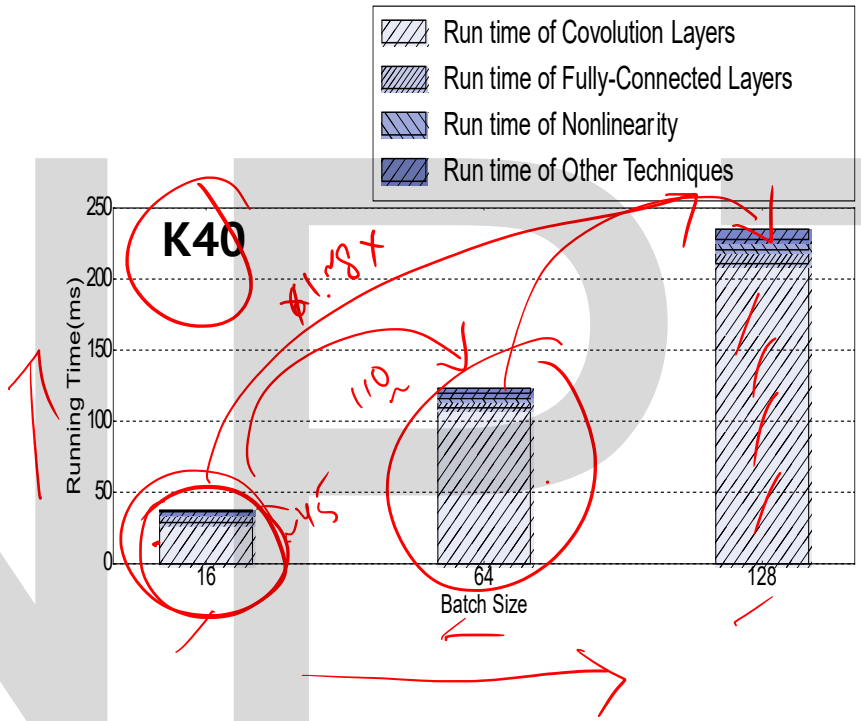


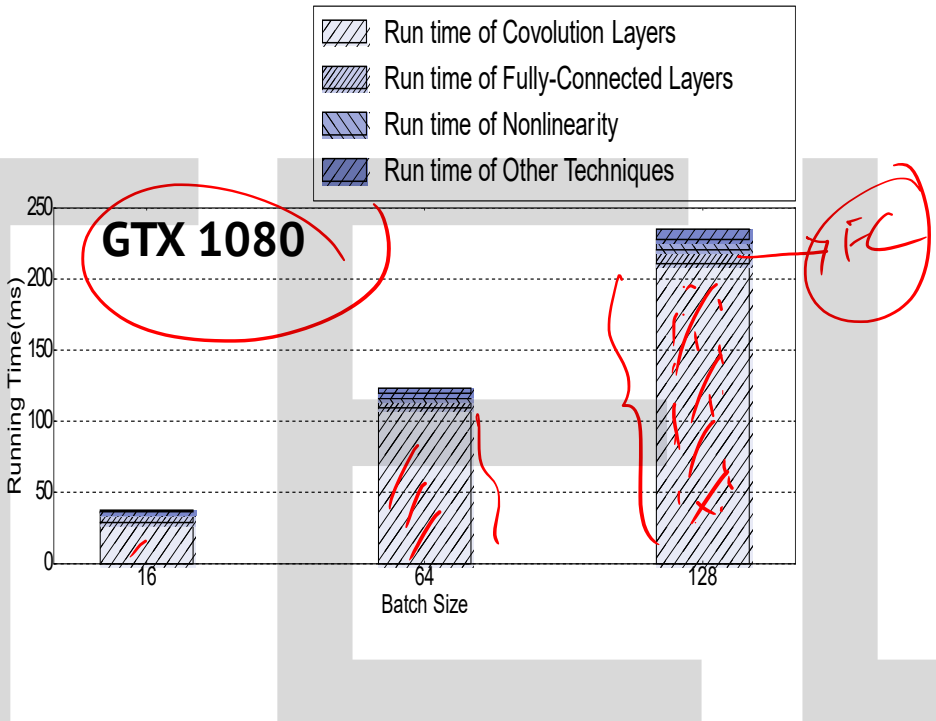
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GPUs Used

Type	Tesla k40	Pascal GTX 1080
Number of processor cores	2880	2560
SIMD lane width	8	
Maximum threads per processor	2048	
Maximum threads per block	1024	
Number of 32-bit registers	65536	
Maximum registers per threads	255	
Shared memory	64KB shared	96KB dedicated
L1 cache	64KB shared	64KB shared
Read-only data cache	48KB dedicated	64KB shared
L2 cache	1536KB	2048KB
GPU maximum clock rate	745Mhz	1607MHz
Memory clock rate	3004Mhz	10000MHz
Memory interface	384-bit	256-bit
Memory bandwidth	208 GB/sec	320 GB/sec
Memory size	12GB	8GB

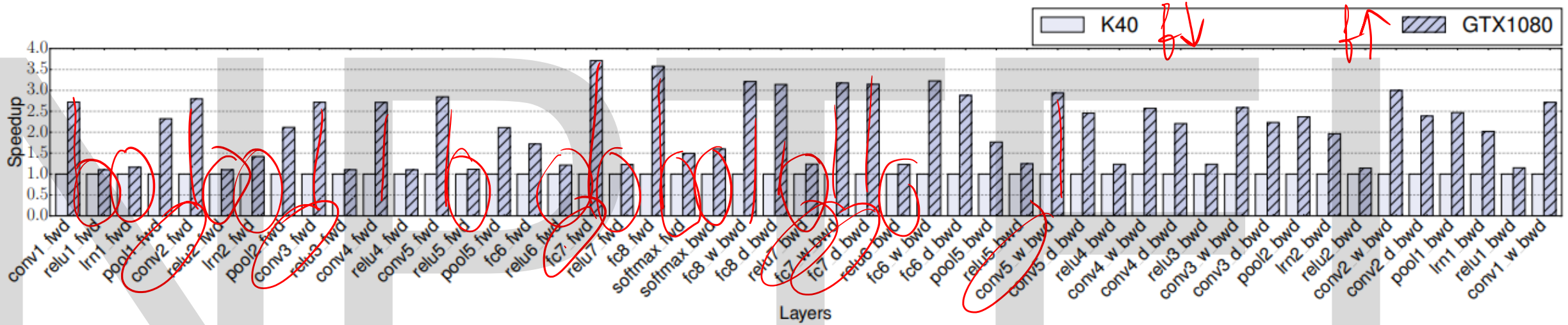
Runtime





- **Runtime dominated by Convolution layers**
- **Larger batch sizes amortize computational cost, especially for convolution layers**

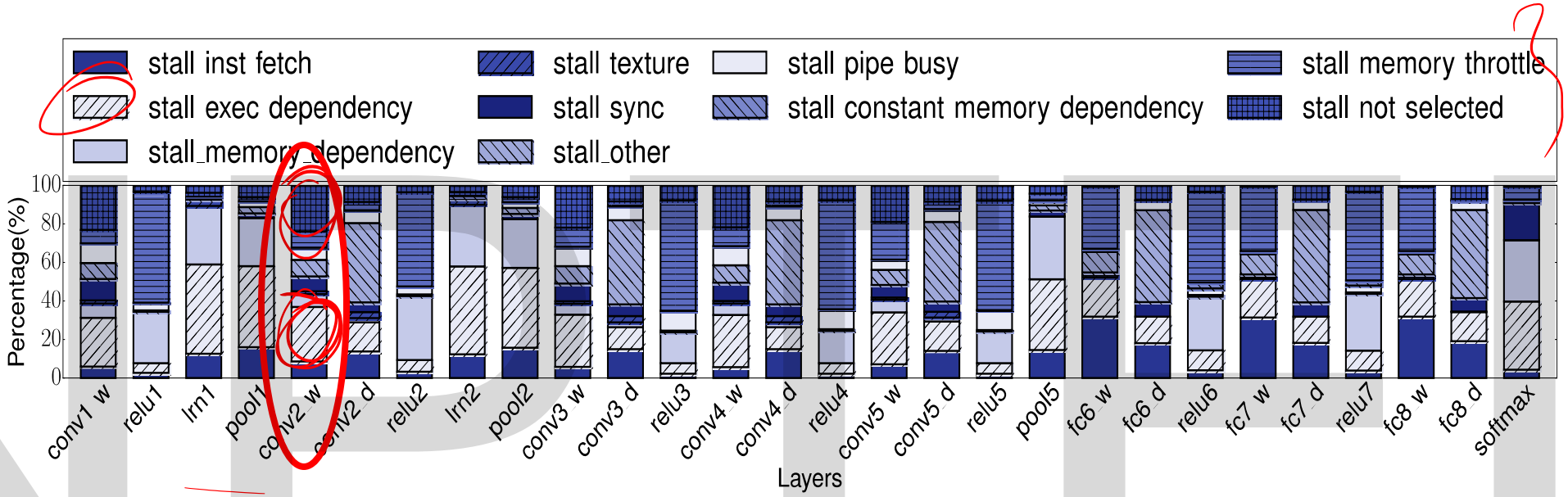
Performance Benefits with Faster GPU



- Normalized with respect to K40
- Higher speedup in Convolution and Fully Connected layers
- Lower benefit in non-linear layers

Understanding the Bottlenecks of Backprop (1)

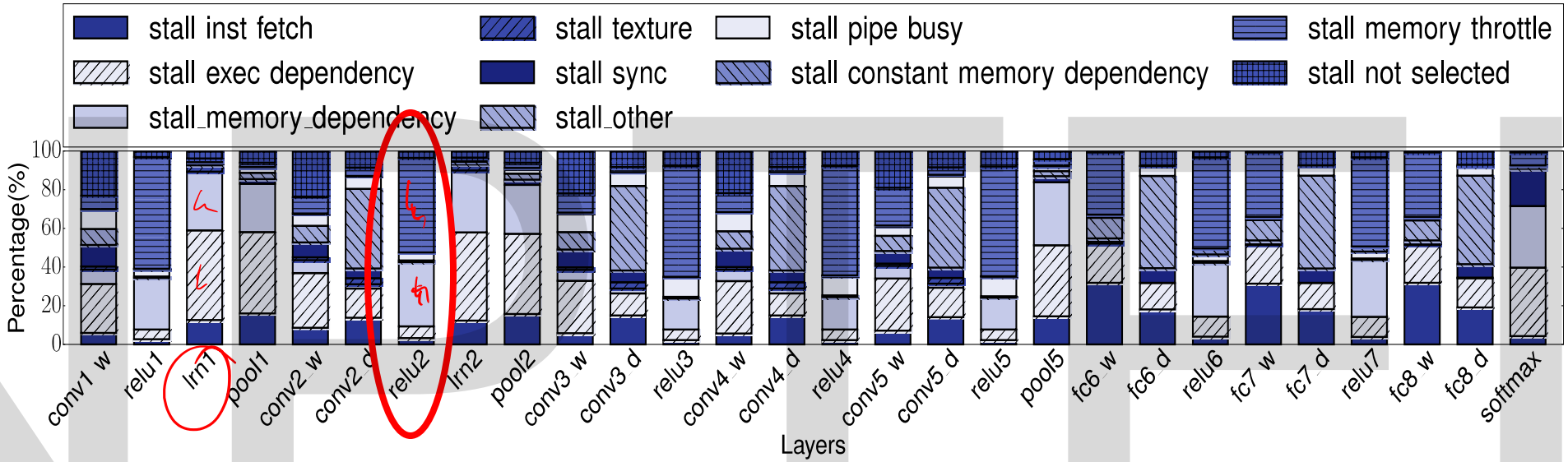
K40



- Two dominating stall categories for conv2_w
  - stall_exec_dependency → many dependencies during instruction execution
  - stall_not_selected → Scheduler selects competing warps.
- conv2_w is mainly bounded by computing.

Understanding the Bottlenecks of Backprop (2)

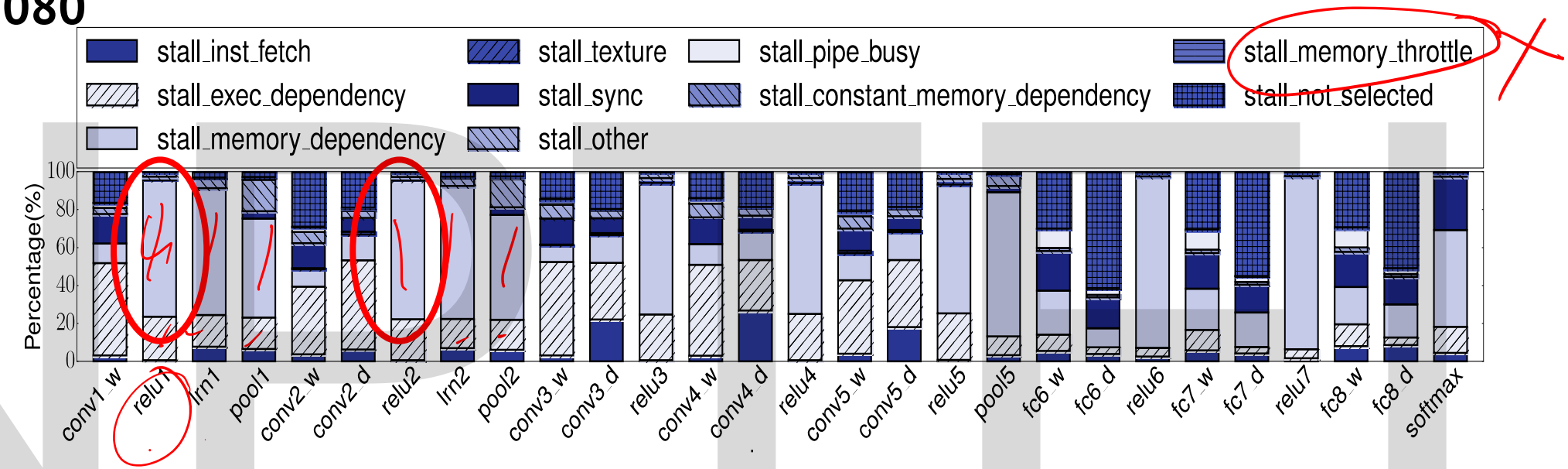
K40



- **Two major stall reasons for relu2 :**
  - **stall_memory_throttle : caused by memory bottlenecks**
  - **stall_memory_dependency : due to program characteristics related to data dependencies on memory loads and stores.**
- **relu2 layer is memory-bound.**
- **lrn2, pool2 & softmax are compute and memory bound**

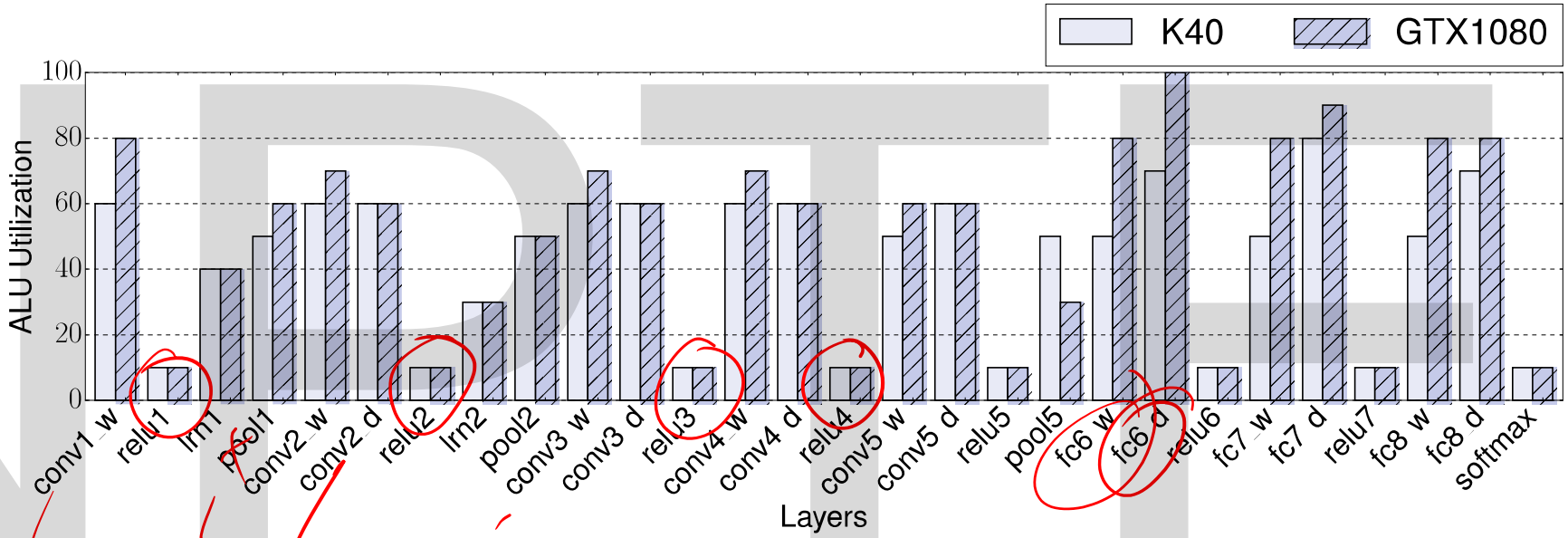
Understanding Bottlenecks of Backprop (3)

GTX1080

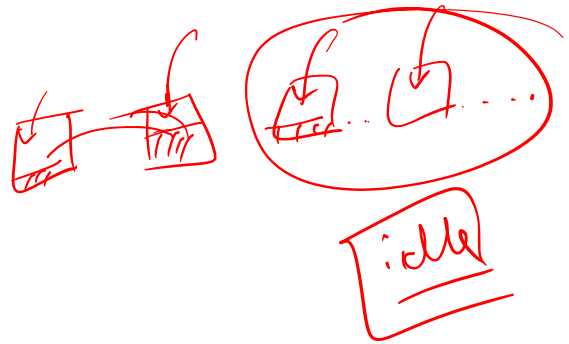


- **Memory stalls contribution is higher in non-linear layers (Relu)**
- **stall_memory_throttle is vanished for all layers**
- **lrn2, pool2 & softmax are memory bound**

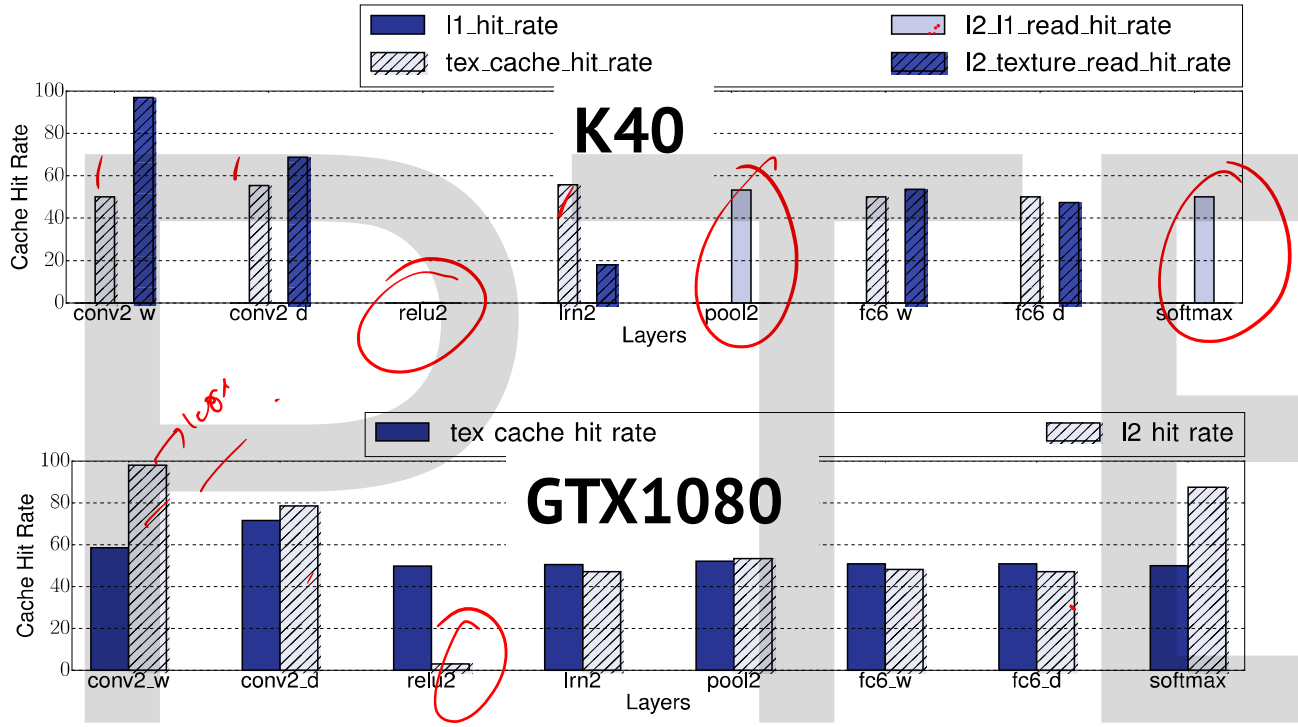
Core Utilization



- **Linear transformation layers have high Core utilization**
- **Non-linear layers (Relu and Softmax) have low utilization**



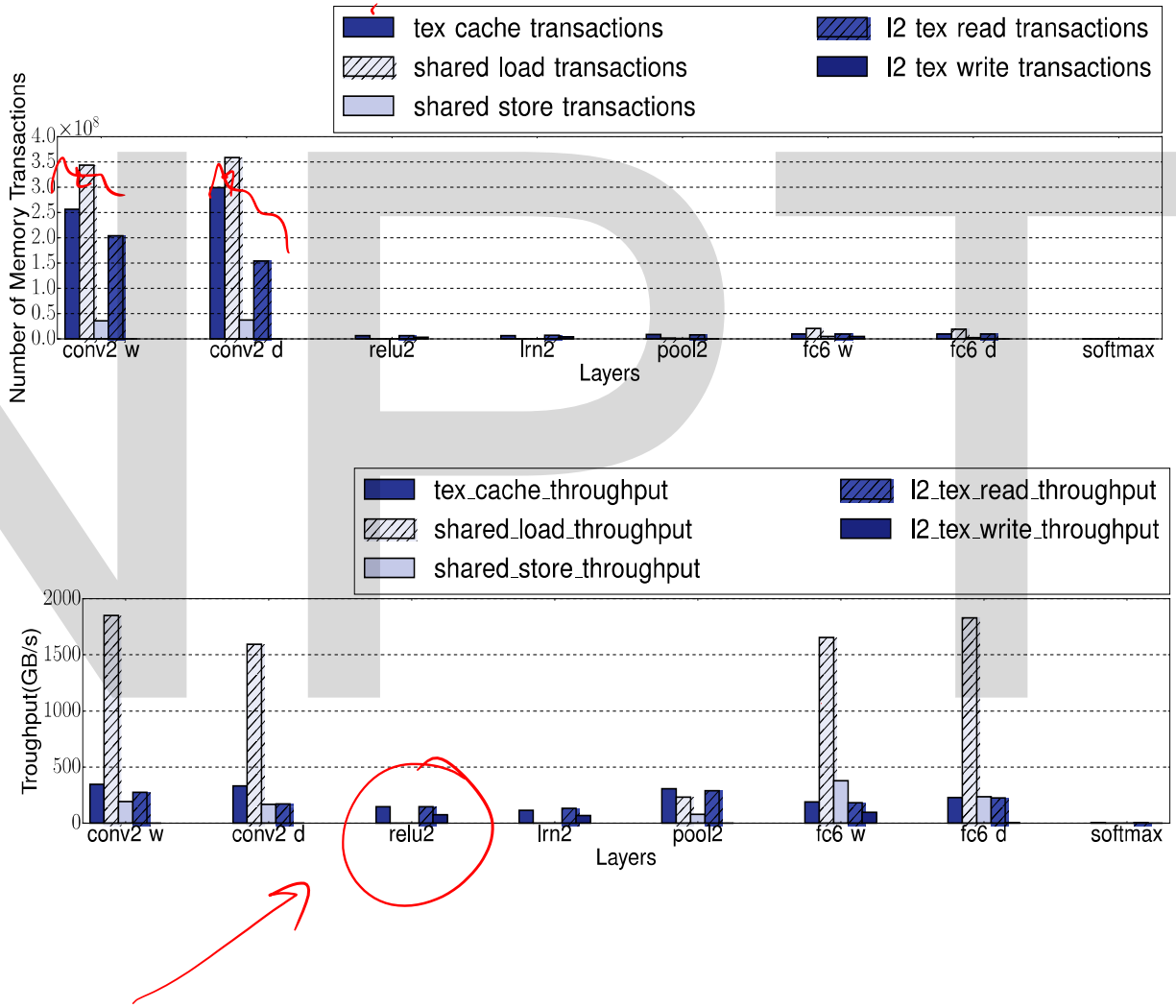
Memory Access Behaviour



- **No observable L1/L2 hit rate in K40.**
- **No L1 hit rate in GTX-1080**
- **Reasonable Texture cache hit**



Memory Transactions



- Higher shared memory in GTX filters access to L2 and DRAM
- Linear layers rely on shared and texture memory
- DRAM throughput in activation layers is higher

Suggested Optimizations

- Increasing the DRAM bandwidth on the GTX1080 will not benefit the CNN throughput.
- Increasing the bandwidth of the texture cache, should result in better performance
- L1 cache is essentially unused; main reason is it is too small to hold data that has strided access.
- Enabling L1 cache bypassing for selected layers (avoids unnecessary data requests to the L1 cache) can help
- Kernel fusion of linear and non-linear layers can help

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THANK YOU

Slide Courtesy: Murali Annavaram (University of Southern California), R. Govindarajan (IISc), Arkaprava Basu (IISc), Jayashree Mohan (Microsoft Research)